ECON 0100 | Exercise D3 | Public Goods

Q1 | The Dumbledore Memorial

Three Dumbledores were considering whether to install a memorial of their brother Albus just outside their town. The individual marginal benefits for three Dumbledores is below.

Aurelius	Aberforth	Ariana
75	10	20

Since the statue will be installed in public space, any of the three are able to enjoy the statue regardless of whether they helped pay for it. It costs 90 coins per year to operate and maintain the memorial.

a) Visit Count

How many of the three Dumbledores would visit the memorial if it were built?

b) Continued Funding

There's an endowment to fund the memorial for its first year. Simply considering their financial interests, how many Dumbledores will contribute to the memorial's continued operation after the endowment runs out?

c) The Referendum

Aurelius proposes a referendum that would raise residents' taxes by 30 coins per year, and puts it to a vote. If each Dumbledore voted in their own private interests, would this referendum receive a majority vote?

d) Marginal Social Benefit

What is the marginal social benefit from the memorial?

e) Efficiency

If the referendum passed, would it lead to an efficient outcome?

Q2 | A Memorial to Dobby

Hogsmeade residents were considering whether to install a memorial of Dobby, the heroic house elf, in their town. The individual marginal benefits for five recently graduated residents is below.

Harry	**Ron**	**Hermione**	**Ginny**	**Neville**
34	29	24	19	14

Any of the five are able to enjoy the statue, regardless of whether they helped pay for it. And it costs 100 coins per year to operate and maintain the memorial.

a) Visit Count

How many of these five residents would visit the memorial if it were built?

b) The Referendum

Neville proposes a referendum that would raise residents' taxes by 20 coins per year, and puts it to a vote. If each of the five Hogsmeade citizens voted in their own private interests, would this referendum receive a majority vote?

c) Marginal Social Benefit

What is the marginal social benefit from the memorial?

d) Efficiency

Would this referendum lead to an efficient outcome?

e) A Higher Tax

Would the proposal receive a majority vote if the tax were 25 coins?

f) Admission

Hermione proposes allowing a private company to build the memorial and charge residents a lump-sum fee to visit the memorial as they like. If the private company were allowed to set a single fee, would the company be able to cover its costs?